

How to Install/Configure and Use SCAR vSomelP stack tool

1. Download and Install :

- Download and install docker compose :

Check docker and "docker compose" installation:

```
docker compose version; docker version
```

If no version is returned for "docker compose" and docker is installed

```
sudo apt remove docker
```

Then execute following steps (from <https://docs.docker.com/engine/install/ubuntu/>)

Remove old versions of Docker tools, those are unofficial and not maintained by Docker, Inc.

```
for pkg in docker.io docker-doc docker-compose docker-compose-v2 podman-docker containerd runc; do sudo apt-get remove $pkg; done
```

Install Docker using the provided convenience script

```
curl -fsSL https://get.docker.com -o install-docker.sh  
sudo sh install-docker.sh
```

Add your user to the docker group to use docker CLI without sudo

```
sudo groupadd docker  
sudo usermod -aG docker $USER
```

Log out and log back in so that your group membership is re-evaluated # Then test that docker is functional

```
docker run hello-world
```

```
cdc-toulouse@cdctoulouse:~/SCAR/scar-master/scripts$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

- **Download SCAR master :**

Download and extract the folder :

```
tar xf scar-master.tar
```

Vbas link Scar master : http://vbas.lge.com:8082/artifactory/renault_cdc/RELEASE/Scar/scar-master.tar

==> Gitlabee Scar master link : https://gitlabee.dt.renault.com/sdv/services/tools/ccs3_integration_tools/scar

==> Scar Readme : [README \(1\).md](#)

```
cdc-toulouse@cdctoulouse:~/SCAR/scar-master$ ls -l
total 92
-rwxrwxr-x 1 cdc-toulouse cdc-toulouse 361 sept. 10 17:23 ci-entripoint.sh
-rw-rw-r-- 1 cdc-toulouse cdc-toulouse 104 sept. 10 17:23 commonapi4someip.ini
-rw-rw-r-- 1 cdc-toulouse cdc-toulouse 569 sept. 10 17:23 compose.yaml
-rw-rw-r-- 1 cdc-toulouse cdc-toulouse 3763 sept. 10 17:23 Dockerfile
drwxrwxr-x 2 cdc-toulouse cdc-toulouse 4096 sept. 10 17:23 docs
drwxrwxr-x 8 cdc-toulouse cdc-toulouse 4096 sept. 10 17:23 ends
drwxrwxr-x 2 cdc-toulouse cdc-toulouse 4096 sept. 10 17:23 grpc
-rw-rw-r-- 1 cdc-toulouse cdc-toulouse 2961 sept. 10 17:23 hardcopy.0
-rw-rw-r-- 1 cdc-toulouse cdc-toulouse 126 sept. 10 17:23 LICENSE
-rw-rw-r-- 1 cdc-toulouse cdc-toulouse 81 sept. 10 17:23 metadata.yml
-rw-rw-r-- 1 cdc-toulouse cdc-toulouse 27499 sept. 10 17:23 README.md
drwxrwxr-x 2 cdc-toulouse cdc-toulouse 4096 sept. 10 17:23 relnotes
-rw-rw-r-- 1 cdc-toulouse cdc-toulouse 630 sept. 10 17:23 requirements_TC.txt
-rw-rw-r-- 1 cdc-toulouse cdc-toulouse 353 sept. 10 17:23 requirements.txt
drwxrwxr-x 9 cdc-toulouse cdc-toulouse 4096 oct. 28 11:51 scripts
drwxrwxr-x 2 cdc-toulouse cdc-toulouse 4096 sept. 10 17:23 sdv_vlan_config
drwxrwxr-x 6 cdc-toulouse cdc-toulouse 4096 sept. 10 17:23 ui
```

- **Download and load Scar Docker Image :**

Download Scar Docker image : http://vbas.lge.com:8082/artifactory/renault_cdc/RELEASE/Scar/scar2.2.0.tar

Then Load the image :

```
sudo docker image load -i scar2.2.0.tar
```

```
cdc-toulouse@cdctoulouse:~/Downloads$ sudo docker image load -i scare1.8.0.tar
e0a9f5911802: Loading layer [=====>] 80.41MB/80.41MB
186313fc554a: Loading layer [=====>] 1.538GB/1.538GB
2f3c8faa7a46: Loading layer [=====>] 783.4kB/783.4kB
b9e2e694638e: Loading layer [=====>] 51.71kB/51.71kB
4c1f4445fa5d: Loading layer [=====>] 121.3MB/121.3MB
7f8a4e011d79: Loading layer [=====>] 1.62MB/1.62MB
f93f4bd2108b: Loading layer [=====>] 48.51MB/48.51MB
6b47ff1479aa: Loading layer [=====>] 49.44MB/49.44MB
65fdada1d24b: Loading layer [=====>] 21.89MB/21.89MB
8881ad0c7ea7: Loading layer [=====>] 7.594MB/7.594MB
6c47c85c0801: Loading layer [=====>] 12.37MB/12.37MB
9ae955fbac8f: Loading layer [=====>] 4.096kB/4.096kB
3c804e72c3a1: Loading layer [=====>] 429.5MB/429.5MB
4bffa32b0bd3: Loading layer [=====>] 757.2kB/757.2kB
c487b2a03d56: Loading layer [=====>] 6.937MB/6.937MB
702ff0a3fc9d: Loading layer [=====>] 6.656kB/6.656kB
0e8f67937a2c: Loading layer [=====>] 6.144kB/6.144kB
b00d0b03c24f: Loading layer [=====>] 6.144kB/6.144kB
1a97e6b6a68f: Loading layer [=====>] 3.072kB/3.072kB
3cbfab5d0c9c: Loading layer [=====>] 3.072kB/3.072kB
f35a65ac471e: Loading layer [=====>] 16.43MB/16.43MB
bde2e9de03a0: Loading layer [=====>] 173.2MB/173.2MB
7c4888541a4f: Loading layer [=====>] 2.168GB/2.168GB
5f70bf18a086: Loading layer [=====>] 1.024kB/1.024kB
69686c71b700: Loading layer [=====>] 20.39MB/20.39MB
Loaded image: artifactory.dt.renault.com/docker-sdv_pmt/sdv/services/tools/ccs3_integration_tools/scar:1.8.0
cdc-toulouse@cdctoulouse:~/Downloads$
```

You can verify if the image is well loaded :

```
cdc-toulouse@cdctoulouse:~/SCAR/scar-master$ docker images
REPOSITORY                                TAG      IMAGE ID       CREATED        SIZE
artifactory.dt.renault.com/docker-sdv_pmt/sdv/services/tools/ccs3_integration_tools/scar  1.8.0    520fbd663889   3 months ago  4.59GB
hello-world                               latest   74cc54e27dc4   9 months ago  10.1kB
```

Confluence install help :<https://confluence.dt.renault.com/spaces/SDVSOL/pages/527308331/vSomeIP+stack+tool++SCAR>

==> PDF extract : [vSomeIP stack tool - SCAR.pdf](#)

Usefull link :

<https://confluence.dt.renault.com/spaces/SDVCAROS/pages/703864503/SCAR+for+CAROS>

==> PDF extract : [SCAR for CAROS_396a273e5c4146ae9454513008cf197c-250925-1322-5578.pdf](#)

<https://confluence.dt.renault.com/spaces/EBI/pages/684769878/Reporting+SCAR>

==> PDF extract : [684769878_e98901ed99b6412baf513d1b9972b8da-250925-1324-5580.pdf](#)

2. Network Configuration :

Download SDV VLAN Network Configuration Tool

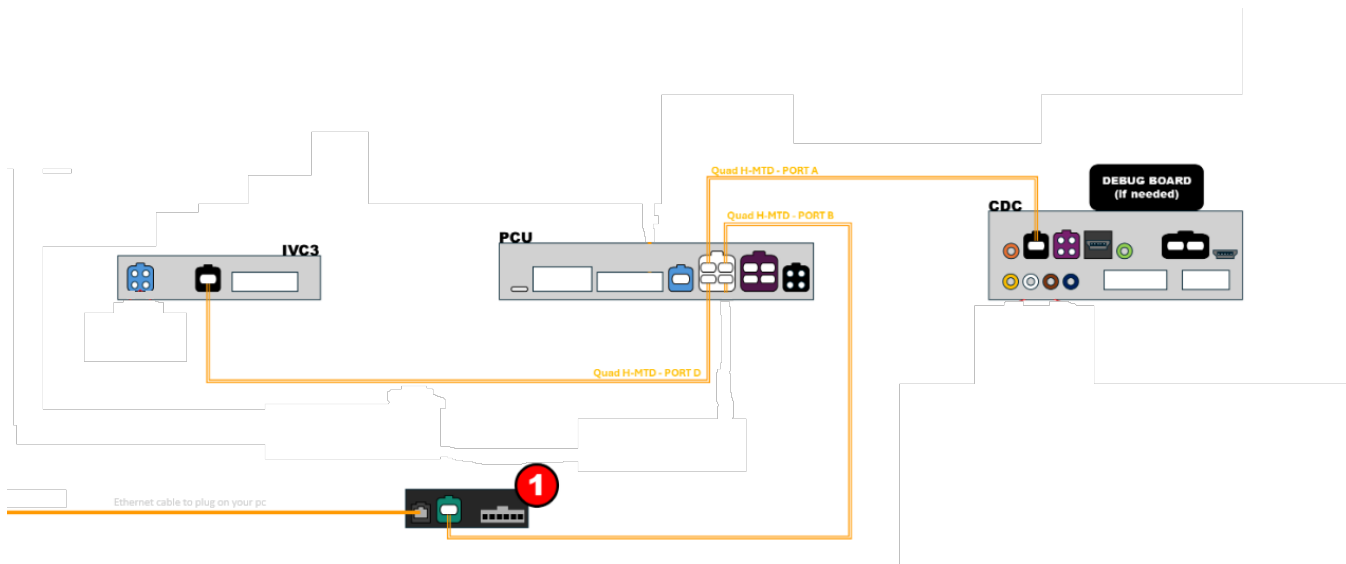
Vbas : http://vbas.lge.com:8082/artifactory/renault_cdc/RELEASE/Scar/sdv_vlan_config-master.tar

Gitlabee : https://gitlabee.dt.renault.com/sdv/services/tools/sdv_vlan_config

First insure that your media converter (Radmoon/Technica) is well plugged and set. Since SCAR is performing only SOME/IP messages, the VLAN configuration must be done before launching any command of SCAR

• Multi Ecu Setup :

if you want to use full Bench with PCU and CDC, you need to plug your converter to the PCU and set the base to 100



First clean your network config :

```
sudo ./netns.sh --setup=hil -i <YOUR_INTERFACE_NAME> --clean
```

Then create a json file in this example it's for **ENDS 2025R10** :

```
{
  "ends": "FaceStep1_2025_R10",
  "ufw": "off",
  "vcans": 0,
  "log_folder": "/tmp",
  "net_config": [
    {
      "interface": "YOUR_INTERFACE_NAME",
      "arp": "on",
      "stub_ecus": [
        "PIU_Mst"
      ]
    }
  ]
}
```

Then run the config :

```
./netns.sh -c multiverse.json
```

```
cdc-toulouse@cdctoulouse:~/SCAR/vlan_config$ cat multiverse.json
{
  "ends": "FaceStep1_2025_R10",
  "ufw": "off",
  "vcans": 0,
  "log_folder": "/tmp",
  "net_config": [
    {
      "interface": "enx74da38fe250a",
      "arp": "on",
      "stub_ecus": [
        "PIU_Mst"
      ]
    }
  ]
}
}cdc-toulouse@cdctoulouse:~/SCAR/vlan_config$ ./netns.sh -c multiverse.json
[sudo] password for cdc-toulouse:
-----
2025-12-11 10:31:40 [DEBUG] MAIN CONFIG ARGS
ends version : FaceStep1_2025_R10
```

- **CDC Standalone :**

If you use CDC standalone you can plug your converter directly to the CDC and set base 1000

First clean your network config :

```
sudo ./netns.sh --setup=hil -i <YOUR_INTERFACE_NAME> --clean
```

Then create standalone.json file in this example it's for **ENDS 2025R10** :

```
{
  "ends": "FaceStep1_2025_R10",
  "ufw": "off",
  "log_folder": "/tmp",
  "net_config": [
    {
      "interface": "YOUR_INTERFACE_NAME",
      "ip": "192.168.1.10",
      "arp": "on",
      "stub_ecus": [
        "PIU_Mst", "PCU_PROXY_FrontEnd", "IVC"
      ],
      "stub_groups": [],
      "conf_type": "veth",
      "cuttlefish": true
    }
  ]
}
```

Then run the config :

```
./netns.sh -c standalone.json
```

3. Usage :

- **to Launch scar**

```
~/scar/scar-master/scripts$ ./scar.sh -t 2.2.0 --ui --arti tls
```

```
cdc-toulouse@cdctoulouse:~/SCAR/scar-master/scripts$ ./scar.sh -t 2.2.0 --ui --arti tls
Using SCAR image from : artifactory.tls.renault.fr
Skip docker pull as requested
[+] Running 1/1
  ✓ Container scar Started0.2s
----- SCAR LAUNCHER -----
Name: scar
Version: 2.2.0
Start : 2025-12-11 10:42:04
Options: {'pull': False, 'tag': '2.2.0', 'debug': False, 'clean': False, 'noentry': False, 'dev': False, 'ui': True, 'arti': 'tls', 'test': '', 'grpc': False, 'clone': False}

!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
You are now into SCAR docker image
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!

No screen session found.
No screen session found.
No Sockets found in /run/screen/S-root.

Validation successful! All button configurations are correct.
capabilities.html generated successfully at /home/scar/ui/.frontend/capabilities.html
☑ HTML file generated: /home/scar/ui/.frontend/ends.html
now open you browser to :
from remote computer (on same network) to scar bench : http://cdctoulouse.tls.renault.fr:8081/ or http://10.202.144.50:8081/
from local scar bench : http://localhost:8081/ or http://127.0.0.1:8081/
root@cdctoulouse:/home/scar#
```

- Check Netns VLAN configuration

check netns can be used in docker

Note : in case you are using VLAN configuration script using netns, to check netns can be used in docker, do following command in docker : ip netns.

in case of CDC standalone, you should see :

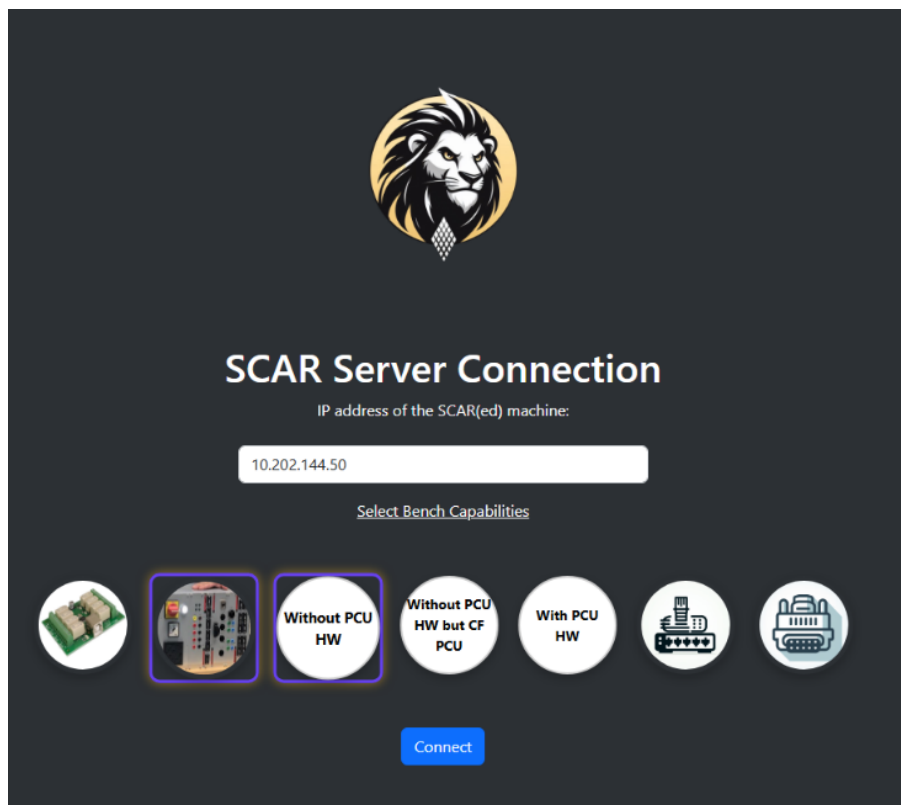
```
root@cdctoulouse:/home/scar# ip netns
PCU_CP_2ns (id: 11)
PCU_CP_1ns (id: 10)
PCU_BC_VM1ns (id: 9)
PCU_PFEs (id: 8)
PCU_DMZ_Fns (id: 7)
PCU_PBEs (id: 6)
PCU_DMZ_Bns (id: 5)
PCU_VISIONns (id: 4)
PIU_Hoodns (id: 3)
PIU_Subns (id: 2)
PIU_Mstns (id: 1)
IVCns (id: 0)
root@cdctoulouse:/home/scar#
```

for multiverse you should see :

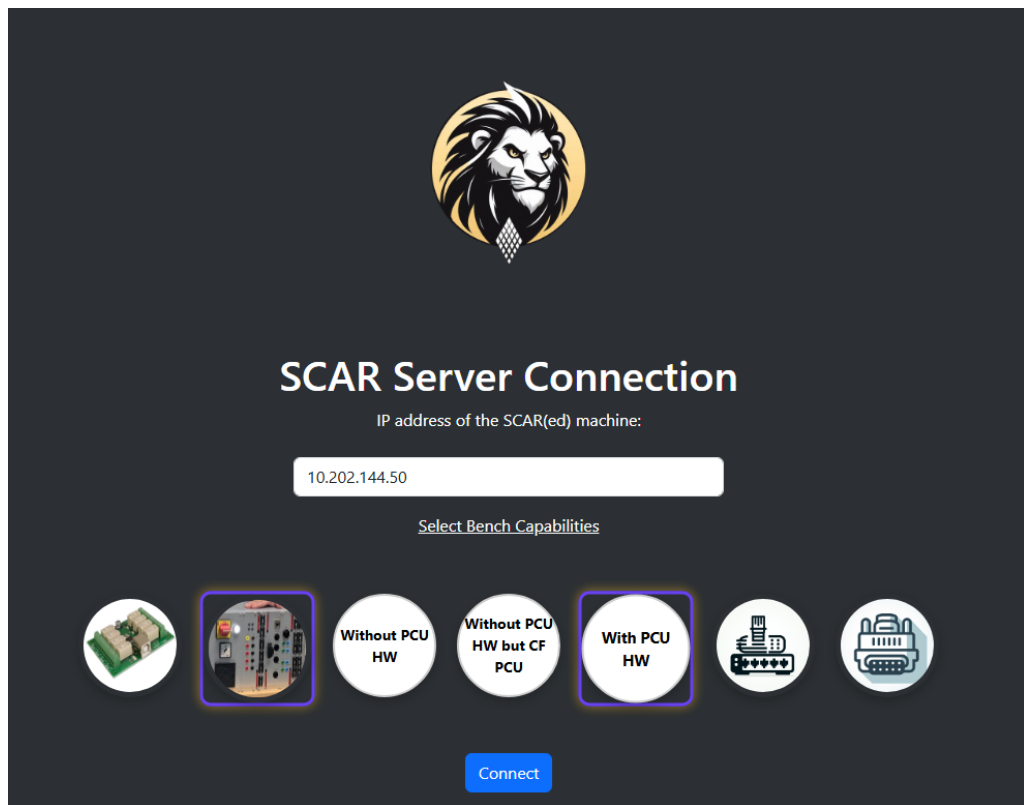
```
root@cdctoulouse:/home/scar# ip netns
PIU_Mstns (id: 1)
root@cdctoulouse:/home/scar#
```

- UI :

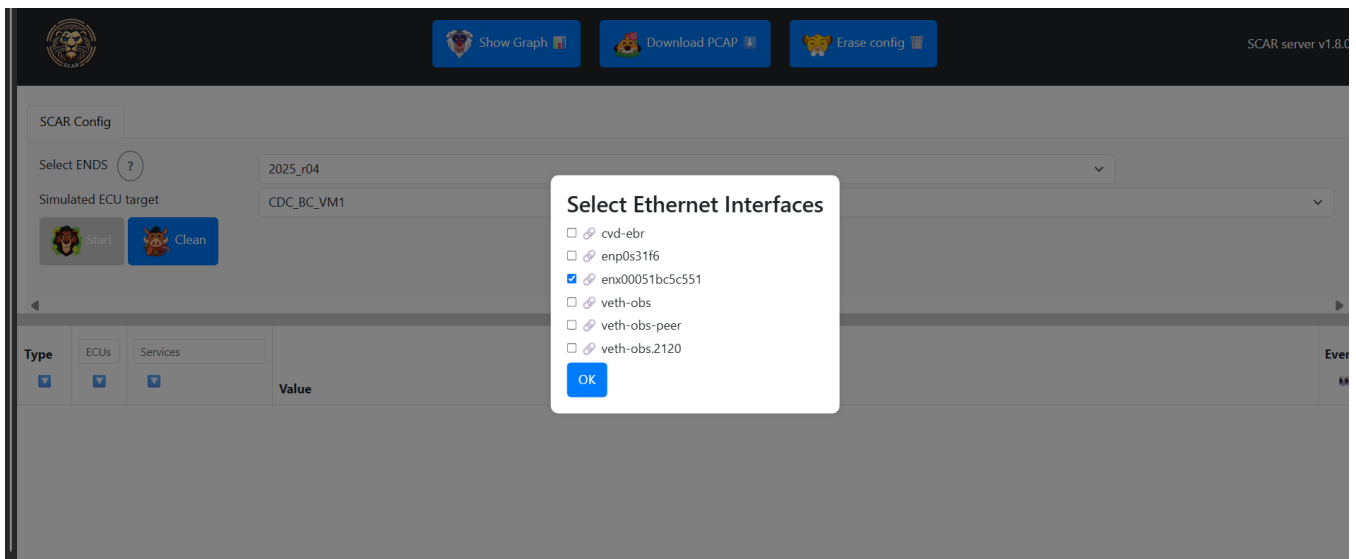
then you can also use the UI : you need to select your configuration, if you are using **CDC standalone** than select this :



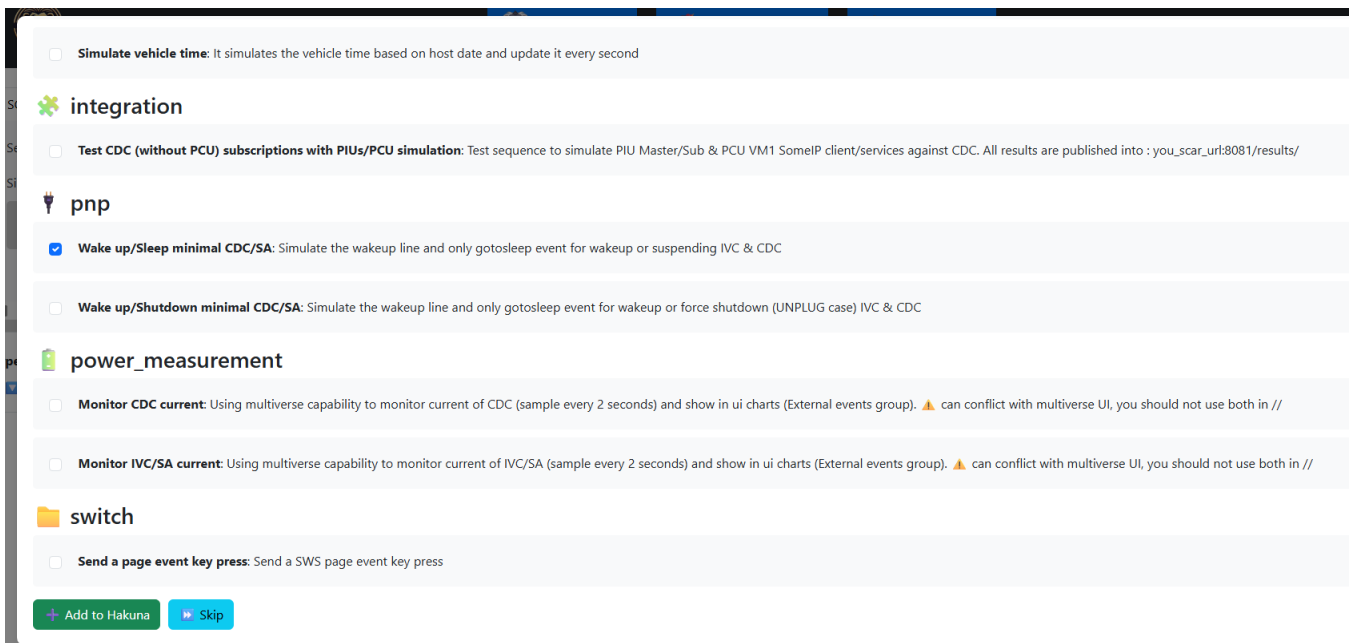
if it's multiverse select this :



then select the interface that you are using for scar :



You can also add button here, select what you need then clic on Add to Hakuna :



- To stop Scar :

```
cdc-toulouse@cdctoulouse:~/SCAR/scar-master/scripts$ ./scar.sh -t 2.2.0 -c
Do you want to clear logs ? [y/n] (default=y) y
Stopping scar docker...
```

4. Set Vehicle Hmi State

Starting from CDC251.02.21 **sdv.cockpit.hmi.state.VehicleHmiState hmi_mode** has to be set to **HMI_MODE_T_MISSION** to have proper displays on RBVM /FVM.

Below you have the steps to follow when using SCAR :

Send HMI MODE using Someip

Start the service using the scar UI

Adjust to match the desired ENDS



 Show Graph
  Download PCAP
  Erase config

SCAR server v1.8.1

SCAR Config

Vehicle State

Select ENDS ?

2025_r05

Simulated ECU target

PIU_Mst

Service to simulate or to register

VehicleHmiState (server)

 Start

 Clean

##Send the signal

```
python3 /home/scar/scripts/test/cmd.py --ends 2025_r05 -s VehicleHmiState -m VehicleHMIState -d hmiMode:HMI_MODE_T_MISSION -d welcomeMode:WELCOME_MODE_T_NOMINAL -d occupantDetected:OCCUPANT_DETECTED_T_ABSENT -d passengerDetected:OCCUPANT_DETECTED_T_ABSENT -d driverDetected:OCCUPANT_DETECTED_T_PRESENT -d isFeedbackNeeded:0
```

```
root@ACTIA-ODC-HHJUX:/home/scar# python3 /home/scar/scripts/test/cmd.py --ends 2025_r05 -s VehicleHmiState -m VehicleHMIState -d hmiMode:HMI_MODE_T_MISSION -d welcomeMode:WELCOME_MODE_T_NOMINAL -d occupantDetected:OCCUPANT_DETECTED_T_ABSENT -d passengerDetected:OCCUPANT_DETECTED_T_ABSENT -d driverDetected:OCCUPANT_DETECTED_T_PRESENT -d isFeedbackNeeded:0
----- CMD -----
Version: 1.8.1
Start : 2025-09-16 12:43:44
Options: {'debug': False, 'service': 'VehicleHmiState', 'message': 'VehicleHMIState', 'period': '', 'data': ['hmiMode:HMI_MODE_T_MISSION', 'welcomeMode:WELCOME_MODE_T_NOMINAL', 'occupantDetected:OCCUPANT_DETECTED_T_ABSENT', 'passengerDetected:OCCUPANT_DETECTED_T_ABSENT', 'driverDetected:OCCUPANT_DETECTED_T_PRESENT', 'isFeedbackNeeded:0'], 'ends': '2025_r05', 'bl': '', 'instance': '0', 'ecu': '', 'action': 'write'}
Log file (host): /home/maroua.chourib@celadodc-rswl.com/scar/scar/scripts/logs/scar/cmd_20250916_12-43-44.log
Log file (docker): /home/scar/scripts/logs/scar/cmd_20250916_12-43-44.log
-----

Please note that write.py and read.py are now available for managing the commands
- hmiMode : HMI_MODE_T_MISSION <-> 5
- welcomeMode : WELCOME_MODE_T_NOMINAL <-> 1
- occupantDetected : OCCUPANT_DETECTED_T_ABSENT <-> 1
- passengerDetected : OCCUPANT_DETECTED_T_ABSENT <-> 1
- driverDetected : OCCUPANT_DETECTED_T_PRESENT <-> 2
- isFeedbackNeeded : 0 <-> False
Update Message VehicleHMIState - ID 33432 for service VehicleHmiStateSvcProv with instance 1 on PIU_Mst using ENDS 2025_r05
Status: Status.SUCCESS
Message: OK
root@ACTIA-ODC-HHJUX:/home/scar#
```

Deep Sleep using SCAR

Configure Hakuna to add the options below



 Show Graph
  Download PCAP
  Erase config

SCAR server v1.6.0

SCAR Config

Bench IO

common

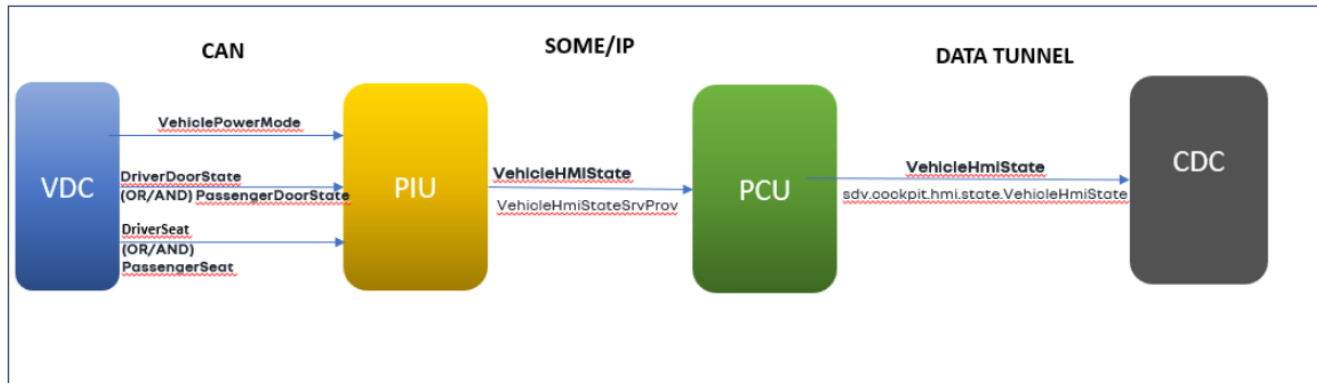
☒ Power supply CDC
 ☒ Power supply Screens CDC
 ☒ Power supply PCU
 ☒ Power supply SA
 ☒ Connection USB HUB

pnp

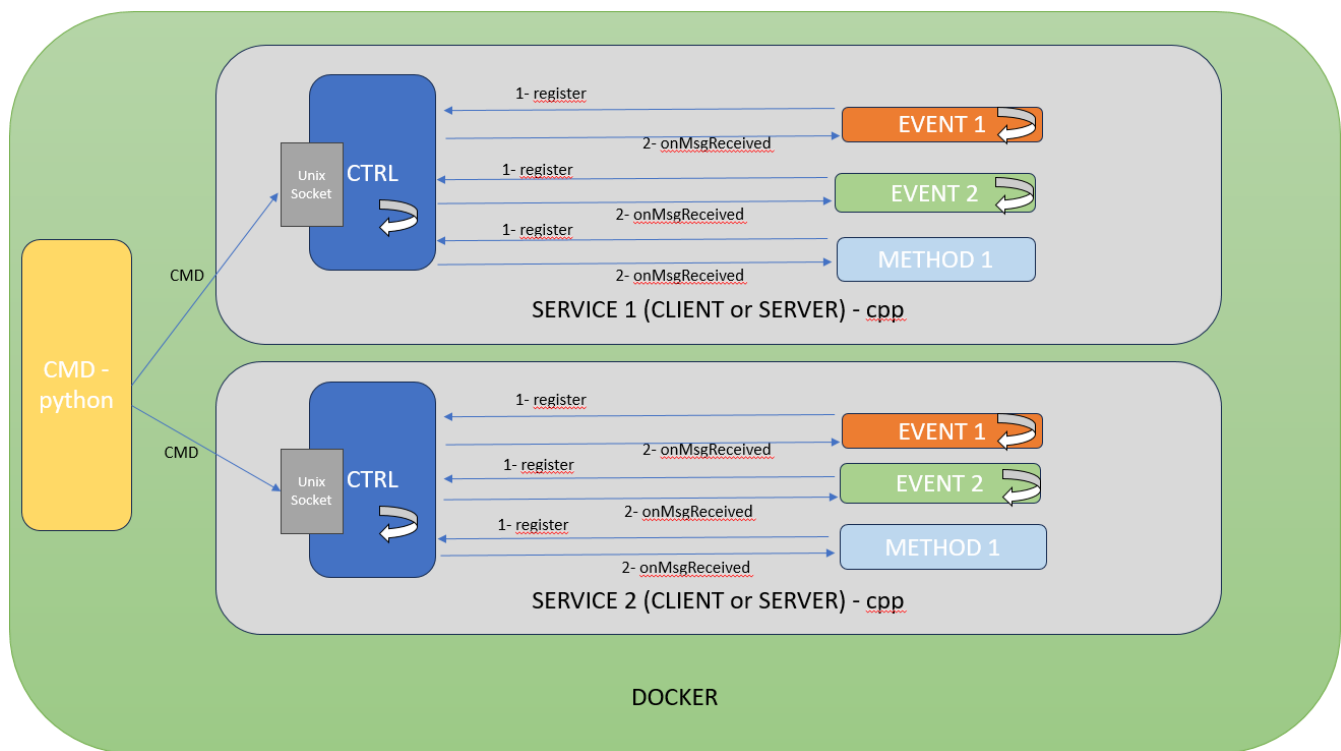
☒ Wake up/Sleep CDC/PCU/SA
 ☐ Wake up/Shutdown CDC/PCU/SA

- For deep sleep: Disable Wake up/sleep CDC/PCU/SA
- For wake up: Enable Wake up/sleep CDC/PCU/SA

General informations :



Scar Architecture :



Vehicle State



<u>VehicleHMIState</u>	<u>VehiclePowerMode</u>
IDLE	LOWER POWER
APPROACH	PARK
ENTRY	PARK
COMFORT	POWER ON
MISSION	TRACTION ON
EXIT	LOWER POWER